

CO3117 - Release baseline diagnostic (WGS-VOA)

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- Machine Learning Foundation:

+ Supervised learning: A machine learning method where model is trained on labeled data to learn mapping function from inputs to outputs.

+ Difference between train, validation and test sets: train set is used directly to fit model parameters and learn underlying parameters weights, validation set used to tune hyperparameter, select the best model checkpoint, and monitor for overfitting, test set: act as an final evaluation set, kept strictly sealed until model selection is frozen to assess generalization on unseen data.

+ Underfitting: the model is too simple to capture the underlying structure of data, overfitting: the model is overly complex and memorize noise patterns in the training data.

+ Bias - Variance: Bias means underfitting: the model can not learn the pattern across train and testing, variance is that model is too sensitive to small fluctuations and noise in the training set.

+ Imagine a learning curve: horizontal axis = complexity of model, vertical axis = error.

When model is too simple: both training and validation error stay close together at high level.

When model is too complex: training error drop near zero, while validation error reach minimum and then increase.

- Decision Tree:

+ Impurity measure the label's percentage distribution between labels. A node contain one class is clean node.

+ Information gain measure the reduction in impurity achieved by partitioning data.

- With the continuous value the decision tree will sort and evaluate a threshold between 2 adjacent values and perform binary splits and choose value that maximizing information gain.

- For missing value, we use mean imputation: replace the missing value in continuous attributes with the training set mean or median.

- Pre-pruning: Stop the tree expansion during training once a stopping criterion meet.

- Post-pruning: allow the tree to grow fully until leaves are ~~pruned~~ pure, then evaluate and remove late